

Concept 4 | Integration: Definite Integrals and Applications

English Student Edition • Focused formulas and guided practice

Formula opener

Antiderivative

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

Definite

$$\int_a^b f(x) dx = F(b) - F(a)$$

Area

$$A = \int_a^b |f(x)| dx$$

Worked Solutions

Integration: Definite Integrals and Applications

Q001 Written

Find the two indefinite integrals in the Warm Up.

Solution

1. Use $\int x^n dx = \frac{x^{n+1}}{n+1} + C$, then differentiate your result to check.

Final answer

Integrate term by term and add C to each primitive.

Q002 Written

Find the six indefinite integrals in the given Try section, expanding a squared bracket first when needed.

Solution

1. For $(x - 1)^2$, write $x^2 - 2x + 1$ before applying the power rule.

Final answer

Expand products or powers, collect like powers, then integrate every term.

Worked Solutions

Integration: Definite Integrals and Applications

Q003 Written

Solve the first ten given indefinite-integral exercises (constants, linear and quadratic polynomials).

Solution

1. A constant integrates to a constant times x ; combine the terms only after integration.

Final answer

Apply linearity and the power rule; retain the constant of integration.

Q004 Written

Solve the remaining eight given integrals, including products and squared brackets.

Solution

1. Do not integrate a product factor-by-factor; expand first, then use the power rule.

Final answer

Expand each product or square, simplify, and integrate term by term.

Worked Solutions

Integration: Definite Integrals and Applications

Q005 **Written****Find $F(x)$ when $F'(x) = 3x^2 + 8x - 1$ and $F(1) = 7$.****Solution**

1. Integrate $F'(x)$, substitute $x = 1$, and solve the resulting equation for C .

Final answer

$$F(x) = x^3 + 4x^2 - x + 3.$$

Q006 **Written****Find the two given Try functions from their derivatives and one given value of F .****Solution**

1. The initial value fixes the otherwise undetermined constant of integration.

Final answerIntegrate the derivative and use the given point to determine C .

Worked Solutions

Integration: Definite Integrals and Applications

Q007 **Written****Find $F(x)$ for all six given Exercise conditions.****Solution**

1. Keep the derivative and the initial condition as two separate equations before solving for C .

Final answerFor each item, integrate $F'(x)$, then impose the stated value of F .Q008 **Written****Find the curve $y = f(x)$ through $(3, 2)$ whose tangent gradient is $3x^2 - 2x + 1$.****Solution**

1. The gradient is $f'(x)$; integrate it and use $f(3) = 2$ to determine C .

Final answer

$$f(x) = x^3 - x^2 + x - 19.$$

Worked Solutions

Integration: Definite Integrals and Applications

Q009 Written

Solve the given Try tangent-gradient problem through (1, 1).**Solution**

1. Replace the gradient by $f'(x)$, integrate, then substitute the known coordinate.

Final answer

Integrate the given gradient and use the point on the curve to determine the constant.

Q010 Written

Solve the three given tangent-gradient exercises, finding each function from its point and gradient.**Solution**

1. A point (x_0, y_0) gives $f(x_0) = y_0$, which fixes C .

Final answer

Integrate the polynomial gradient and apply the given point condition in each case.

Worked Solutions

Integration: Definite Integrals and Applications

Q011 Written

Evaluate the three definite integrals in the Warm Up using the Fundamental Theorem of Calculus.

Solution

1. Write $[F(x)]_a^b = F(b) - F(a)$ before substituting the limits.

Final answer

Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero.

Q012 Written

Evaluate the six given Try definite integrals, simplifying the integrand before applying the limits.

Solution

1. Linearity allows sums and differences of integrals to be combined before integration.

Final answer

Expand products, integrate, and evaluate the antiderivative at the upper and lower limits.

Worked Solutions

Integration: Definite Integrals and Applications

Q013 Written

Evaluate the first twelve given Exercise definite integrals, including reversed and equal limits.

Solution

1. Check the order of the limits before evaluating the antiderivative.

Final answer

Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0.

Q014 Written

Evaluate the remaining twelve given Exercise integrals, including sums, differences and products.

Solution

1. Combine only integrals with the same limits; otherwise split or evaluate separately.

Final answer

Use linearity, expand products, then apply the endpoint-substitution rule.

Worked Solutions

Integration: Definite Integrals and Applications

Q015 Written

Use the four properties of definite integrals to simplify the three Warm Up expressions.

Solution

1. Choose the property that makes the integrand or limits match a known form.

Final answer

Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate.

Q016 Written

Solve the four given Try definite-integral transformations.

Solution

1. For $(x - a)(x - b)$, first make the coefficient of x equal to 1.

Final answer

Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity.

Worked Solutions

Integration: Definite Integrals and Applications

Q017 **Written****Solve the first six given Exercise questions using interval splitting and reversal.****Solution**

1. A minus sign before an integral can be absorbed by reversing its limits.

Final answer

Rewrite each expression until adjacent intervals have matching integrands and compatible limits.

Q018 **Written****Solve the final six given Exercise questions using the factorized definite-integral identity.****Solution**

1. Identify (a,b) from the two linear factors and account for every constant multiplier.

Final answerScale to $\int_a^b (x - a)(x - b) dx = -\frac{(b-a)^3}{6}$.

Worked Solutions

Integration: Definite Integrals and Applications

Q019 Written

Find $f(x)$ from the given equation $f(x) = x^2 + \int_0^1 f(t) dt$.

Solution

1. Replace the definite integral by a constant, substitute the resulting $f(t)$ back into it, and solve.

Final answer

Let $A = \int_0^1 f(t) dt$; then $A = \frac{1}{3} + A$, so $A = -\frac{1}{3}$ and $f(x) = x^2 - \frac{1}{3}$.

Q020 Written

Solve the two given Try integral equations for $f(x)$.

Solution

1. The unknown integral is a number independent of x ; determine it before writing the final function.

Final answer

Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation.

Worked Solutions

Integration: Definite Integrals and Applications

Q021 **Written****Find $f(x)$ for all five given integral equations.****Solution**

1. Check the final function by integrating it over the stated interval and substituting back.

Final answer

Let the definite integral be A , substitute the expression for $f(t)$, solve for A , then recover $f(x)$.

Q022 **Written****Differentiate $\int_3^x (2t^2 + 3t - 5) dt$ with respect to x .****Solution**

1. By the Fundamental Theorem, $\frac{d}{dx} \int_a^x f(t) dt = f(x)$.

Final answer

$2x^2 + 3x - 5$.

Worked Solutions

Integration: Definite Integrals and Applications

Q023 Written

Solve the two given Try problems involving differentiation of a variable-limit integral and an unknown lower limit.

Solution

1. A constant lower limit disappears under differentiation; it is recovered from the original equation.

Final answer

Replace t by x when differentiating, then use equal limits or the given condition to determine a .

Q024 Written

Differentiate the three given expressions with respect to x .

Solution

1. After differentiation, replace t with the variable upper limit x .

Final answer

Apply the variable-limit Fundamental Theorem directly to each integrand.

Worked Solutions

Integration: Definite Integrals and Applications

Q025 **Written****Find $f(x)$ and a for the three given equations with a variable upper limit.****Solution**

1. Use $0 = \int_a^x f(t) dt$ in the original equation after finding $f(x)$.

Final answer

Differentiate to get $f(x)$, then set the integral to zero at $x = a$ to solve for a .

Q026 **MCQ****The constant of integration is required because:****Solution**

1. Differentiation removes additive constants.

Correct option: B**Final answer**

primitives differ by a constant

Worked Solutions

Integration: Definite Integrals and Applications

Q027 MCQ

The integral of x^n for $n \neq -1$ is:**Solution**

1. Use the power rule for integration.

Correct option: B

Final answer

$$x^{(n+1)/(n+1)} + C$$

Q028 MCQ

Before integrating $(x - 1)^2$, first:**Solution**1. Write $x^2 - 2x + 1$.

Correct option: B

Final answer

expand it

Worked Solutions

Integration: Definite Integrals and Applications

Q029 MCQ

The integral of $3x^2 - 2x + 1$ is:**Solution**

1. Integrate term by term.

Correct option: A**Final answer**

$$x^3 - x^2 + x + C$$

Q030 MCQ

To find $F(x)$ from $F'(x)$, you should first:**Solution**

1. Integration reverses differentiation.

Correct option: B**Final answer**integrate $F'x$

Worked Solutions

Integration: Definite Integrals and Applications

Q031 MCQ

The condition $F(1) = 7$ determines:**Solution**

1. Use the given function value after integrating.

Correct option: C

Final answer

C

Q032 MCQ

If $F'(x) = 6x - 2$ and $F(0) = 1$, then $F(x)$ is:**Solution**1. Integrate and use $F(0) = 1$.

Correct option: A

Final answer $3x^2 - 2x + 1$

Worked Solutions

Integration: Definite Integrals and Applications

Q033 MCQ

The gradient of the tangent at (x, y) is:**Solution**

1. By definition, the derivative is the tangent gradient.

Correct option: B**Final answer**

$$f'x$$

Q034 MCQ

A point (x_0, y_0) on $y = f(x)$ gives:**Solution**

1. Use the point as an initial condition.

Correct option: B**Final answer**

$$f(x_0) = y_0$$

Worked Solutions

Integration: Definite Integrals and Applications

Q035 MCQ

If $f'(x) = 3x^2 - 2x + 1$, an antiderivative is:

Solution

1. Integrate each term.

Correct option: B

Final answer

$$x^3 - x^2 + x + C$$

Q036 MCQ

The Fundamental Theorem gives $\int_a^b f(x)dx =$:

Solution

1. Evaluate an antiderivative at the upper minus lower limit.

Correct option: B

Final answer

$$Fb - Fa$$

Worked Solutions

Integration: Definite Integrals and Applications

Q037 MCQ

When the limits are equal, $\int_a^a f(x)dx$ equals:

Solution

1. There is no interval of accumulation.

Correct option: C

Final answer

0

Q038 MCQ

A reversed definite integral satisfies:

Solution

1. Reversing the limits changes the sign.

Correct option: B

Final answer

negative reversed integral

Worked Solutions

Integration: Definite Integrals and Applications

Q039 MCQ

To evaluate a definite integral of a product, first:**Solution**

1. Then apply the power rule and endpoint substitution.

Correct option: B

Final answer

expand the product

Q040 MCQ

Adjacent integrals with the same integrand can be:**Solution**

1. Use interval additivity.

Correct option: B

Final answer

combined over the joined interval

Worked Solutions

Integration: Definite Integrals and Applications

Q041 MCQ

To use $\int_{\alpha}^{\beta} (x - \alpha)(x - \beta)dx$, the factors must have:

Solution

1. Match each factor to its endpoint.

Correct option: B

Final answer

roots a and b

Q042 MCQ

The identity for the factored integral is:

Solution

1. Use the signed cubic interval length.

Correct option: B

Final answer

$-(b - a)^3/6$

Worked Solutions

Integration: Definite Integrals and Applications

Q043 MCQ

A minus sign before an integral can be removed by:

Solution

1. $-\int_a^b f = \int_b^a f.$

Correct option: B

Final answer

reversing its limits

Q044 MCQ

In $f(x) = p(x) + \int_a^b f(t)dt$, the definite integral is:

Solution

1. The limits contain no x .

Correct option: B

Final answer

a constant

Worked Solutions

Integration: Definite Integrals and Applications

Q045 MCQ

The best first step for an integral equation is to let:**Solution**

1. Replace the repeated definite integral by one symbol.

Correct option: B**Final answer** $A = \text{integral}$

Q046 MCQ

For the warm-up $f(x) = x^2 + \int_0^1 f(t)dt$, the constant integral is:**Solution**

1. Solve $A = 1/3 + A$ after substitution.

Correct option: B**Final answer** $-1/3$

Worked Solutions

Integration: Definite Integrals and Applications

Q047 MCQ

$\frac{d}{dx} \int_a^x f(t)dt$ equals:

Solution

1. Apply the variable-upper-limit theorem.

Correct option: B

Final answer $f(x)$

Q048 MCQ

When differentiating an integral, the dummy variable t is replaced by:

Solution

1. The derivative evaluates the integrand at the moving limit.

Correct option: B

Final answer

upper-limit variable

Worked Solutions

Integration: Definite Integrals and Applications

Q049 MCQ

If $\int_a^a f(t)dt = 0$, this can determine:

Solution

1. Substitute the equal limits into the original equation.

Correct option: B

Final answer

the unknown lower limit a

Q050 MCQ

For $\int_3^x (2t^2 + 3t - 5)dt$, the derivative is:

Solution

1. Use the Fundamental Theorem directly.

Correct option: A

Final answer

$2x^2 + 3x - 5$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz WRITTEN 1 Concept Quiz · Written

Find $\int (4x^3 - 6x + 2)dx$.

Solution

1. Apply the power rule term by term.

Final answer

$$x^4 - 3x^2 + 2x + C$$

Quiz WRITTEN 2 Concept Quiz · Written

Find $F(x)$ if $F'(x) = 2x - 5$ and $F(2) = 1$.

Solution

1. Integrate and use $F(2) = 1$.

Final answer

$$F(x) = x^2 - 5x + 7$$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz WRITTEN 3 Concept Quiz · Written

Evaluate $\int_{-1}^2 (3x^2 - 4x + 1)dx$.

Solution

1. Use an antiderivative and subtract the endpoint values.

Final answer

6

Quiz WRITTEN 4 Concept Quiz · Written

Find $f(x)$ if $f(x) = x^2 + \int_0^1 f(t)dt$.

Solution

1. Let $A = \int_0^1 f(t)dt$; solve $A = 1/3 + A$ after substitution.

Final answer

$f(x) = x^2 - 1/3$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz MCQ 5 [Concept Quiz · MCQ](#)

The derivative of $\int_0^x (3t^2 + 1)dt$ is:

Solution

1. Evaluate the integrand at $t = x$.

Correct option: A

Final answer

$$3x^2 + 1$$

Quiz MCQ 6 [Concept Quiz · MCQ](#)

A primitive of $5x^4$ is:

Solution

1. Divide the power coefficient by the new exponent.

Correct option: A

Final answer

$$x^5 + C$$

Worked Solutions

Integration: Definite Integrals and Applications

Quiz MCQ 7 Concept Quiz · MCQ

For $F'(x) = 4x + 2$ and $F(0) = 3$, $F(x)$ is:

Solution

1. Integrate and apply $F(0) = 3$.

Correct option: A

Final answer

$$2x^2 + 2x + 3$$

Quiz MCQ 8 Concept Quiz · MCQ

$\int_1^3 f(x)dx + \int_3^5 f(x)dx$ equals:

Solution

1. Join adjacent intervals with the same integrand.

Correct option: A

Final answer

integral from 1 to 5

Answer key

Use the question number shown on each page.

- | | | | |
|---|---|---|--------------------------------------|
| Q001 Integrate term by term and add C to each primitive. | Q012 Expand products, integrate, and evaluate the antiderivative at the upper and lower limits. | Q020 Introduce a constant for the definite integral, evaluate it using the proposed $f(t)$, and solve the linear equation. | Q036 B |
| Q002 Expand products or powers, collect like powers, then integrate every term. | Q013 Use the Fundamental Theorem; a reversed interval changes the sign and equal limits give 0. | Q021 Let the definite integral be A , substitute the expression for $f(t)$, solve for A , then recover $f(x)$. | Q037 C |
| Q003 Apply linearity and the power rule; retain the constant of integration. | Q014 Use linearity, expand products, then apply the endpoint-substitution rule. | Q022 $2x^2 + 3x - 5$. | Q038 B |
| Q004 Expand each product or square, simplify, and integrate term by term. | Q015 Scale constants, reverse limits, split intervals, or use the factored-interval identity as appropriate. | Q023 Replace t by x when differentiating, then use equal limits or the given condition to determine a . | Q039 B |
| Q005 $F(x) = x^3 + 4x^2 - x + 3$. | Q016 Swap limits when necessary, join adjacent intervals, and normalize a linear factor before using the identity. | Q024 Apply the variable-limit Fundamental Theorem directly to each integrand. | Q040 B |
| Q006 Integrate the derivative and use the given point to determine C . | Q017 Rewrite each expression until adjacent intervals have matching integrands and compatible limits. | Q025 Differentiate to get $f(x)$, then set the integral to zero at $x = a$ to solve for a . | Q041 B |
| Q007 For each item, integrate $F'(x)$, then impose the stated value of F . | Q018 Scale to $\int_a^b (x-a)(x-b) dx = -\frac{(b-a)^3}{6}$. | Q026 B | Q042 B |
| Q008 $f(x) = x^3 - x^2 + x - 19$. | Q019 Let $A = \int_0^1 f(t) dt$; then $A = \frac{1}{3} + A$, so $A = -\frac{1}{3}$ and $f(x) = x^2 - \frac{1}{3}$. | Q027 B | Q043 B |
| Q009 Integrate the given gradient and use the point on the curve to determine the constant. | | Q028 B | Q044 B |
| Q010 Integrate the polynomial gradient and apply the given point condition in each case. | | Q029 A | Q045 B |
| Q011 Find an antiderivative F and calculate $F(b) - F(a)$; equal limits give zero. | | Q030 B | Q046 B |
| | | Q031 C | Q047 B |
| | | Q032 A | Q048 B |
| | | Q033 B | Q049 B |
| | | Q034 B | Q050 A |
| | | Q035 B | Quiz WRITTEN 1 $x^4 - 3x^2 + 2x + C$ |
| | | | Quiz WRITTEN 2 $F(x) = x^2 - 5x + 7$ |
| | | | Quiz WRITTEN 3 6 |
| | | | Quiz WRITTEN 4 $f(x) = x^2 - 1/3$ |
| | | | Quiz MCQ 5 A |
| | | | Quiz MCQ 6 A |
| | | | Quiz MCQ 7 A |
| | | | Quiz MCQ 8 A |